

Muhammad Ahsan Zahoor

Software Engineer

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🔗 github.com/codewithahsanzahoor 🔗 ahsan-zahoor-portfolio.vercel.app

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Professional Experience

Associate Software Engineer - 1 year, Preesoft

06/2025 – 07/2026

Lahore, Pakistan

1. *Cognify / Brighton AI Education Platform:*

- Contributed to a **multi-tenant Learning Management System (LMS)** with role-based dashboards for Admins, Teachers, Students, and Parents.
- Worked on **both frontend and backend** in a team of 20+ developers using React, TypeScript, Node.js, Express, MongoDB, and microservices.
- Implemented/handled features involving **JWT authentication, Stripe subscriptions, AWS S3 storage, role-based access control, and secure inter-service communication.**
- Worked with **Figma-to-code implementation** using Tailwind CSS, SCSS, Shadcn UI, Material UI, and reusable global components.
- Contributed to modules including **AI-powered Exam Builder, academic management, user management, storage, and monetization.**

2. *Bookify - <https://bvh.preesoft.net/dashboard>:* 🔗

- Worked as a **Full-Stack Developer** on a multi-tenant SaaS **Billing & Organization Management system.**
- Contributed to backend development using **Node.js, TypeScript, Express, Prisma ORM, MySQL, and Dependency Injection with Inversify.**
- Developed/maintained dashboard features using **React 19, Vite, Material UI v7, and SWR** for efficient data fetching and caching.
- Implemented/handled **JWT authentication, Role-Based Access Control (RBAC), route guards, reusable components, and type-safe form validation using React Hook Form and Zod.**
- Worked in a small backend team of only **2 developers**, giving hands-on full-stack and backend experience.

3. *Talent Bridge - <https://talent-dev.blackhives.com/>:* 🔗

- Worked as a **Frontend Developer** on a multi-tenant Job Board and Learning Management System.
- Built frontend features using **Next.js App Router, React 19, TypeScript, Redux Toolkit, and Tailwind CSS.**
- Implemented **JWT silent token refresh** using Axios interceptors, including request queuing to prevent duplicate refresh calls.
- Developed **authentication and role-based permission systems** using AuthGuard, GuestGuard, RoleBasedGuard, and granular permissions.
- In this, I only worked on frontend. In this, there were 10 team members, full stack, frontend and backend.

Javascript Developer - 1 year, Rizviz International Impex

05/2024 – 04/2025

Lahore, Pakistan

- Developed custom ServiceNow Service Portal components using HTML, CSS, and JavaScript.
- Implemented client-side and server-side JavaScript using **Client Scripts, Business Rules, and Script Includes** to automate business logic and repetitive tasks.
- Configured **Catalog Items, Record Producers, UI Policies, and Data Policies** to improve internal workflows and form behavior.
- Built process automations using **Flow Designer**, including approvals and task assignments.
- Integrated **REST APIs** to exchange data between ServiceNow and external systems.
- Debugged and resolved issues using ServiceNow logs and debugging tools.

Freelance Full Stack Developer / Independent Developer - 1.5 year

11/2022 – 06/2024

Remote

- Designed, developed, and deployed scalable web applications using the **MERN stack (MongoDB, Express.js, React, and Node.js)** based on client requirements.
- Developed responsive and high-performance frontend applications using **Next.js, React, TypeScript, and Tailwind CSS**, converting UI designs into pixel-perfect interfaces.
- Engineered robust backend architectures and developed efficient **RESTful APIs** for secure and reliable communication between frontend and backend systems.
- Designed database schemas and implemented data management solutions using **MongoDB and Mongoose**.
- Implemented authentication and authorization systems using **JWT-based access and refresh tokens**.
- Built reusable, maintainable UI components and managed application state using **Zustand and Context API**.
- Utilized **Git and GitHub** for version control, maintaining clean, organized, and well-documented code repositories.
- Debugged, optimized, and deployed applications while collaborating with clients to understand requirements and deliver customized software solutions.

Final Year Project - Medical Document OCR & Text Extraction System,

https://github.com/codewithahsanzaheer/fyp_ocr_medical_detail 

- **Single-handedly completed my FYP**—merging academic research with hands-on development to deliver a fully functional project and dissertation.
- Developed a **Medical Document OCR and Text Extraction System** to extract structured information from scanned medical documents.
- Built an OCR pipeline using **FastAPI, Streamlit, OpenCV, Tesseract OCR, pytesseract, and Regular Expressions**.
- Implemented the pipeline: **PDF → Images → Image Preprocessing → OCR → Regex Parsing → Structured JSON**.
- Applied **grayscale conversion, image resizing, and adaptive thresholding** to improve OCR accuracy on noisy scanned documents.
- Added additional features including **handwritten OCR, text translation/summarization using Hugging Face Transformers, and medicine information lookup**.

Education

Bachelors of Computer Science, Government College University

2020 – 2024

- GPA: 3.41/4

Lahore, Pakistan

Skills

AI & Automation (Competent)

Gemini CLI, Cursor AI, Prompt Engineering, Antigravity, Stitch, Copilot and Copilot Cli, Vercel, Qwen, Devin Ai

Reactjs, Nextjs with tailwindcss (Proficient)

I use these to build the front end of all my apps, from a full website that is standard in the industry to a small landing page. I also use react on the mobile end side now with reactnative cli.

mysql, postgres with prisma (Amateur)

I used it to build the backend of my company project called Bookify. There, I learned how to use these databases with Prisma.

TypeScript (Competent)

I now use it a lot when I build the front or back end of my apps in JavaScript because it's easy to debug.

PHP (Amateur)

I used it on a landing page and also on a four-page website with a MySQL database for submitting forms.

NestJs (Amateur)

I used it to develop the backend of bookify project, and for the DB, I used prisma with postgres db.

Javascript (Proficient)

I used it to make apps on both the front end and the back end.

Python (Competent)

I used it to learn about programming in general and to make a Jarvis app. I also used it to make the back end of my apps with FastAPI and Django.

DevOps & Tools (Competent)

Docker (for containerization), Linux (as my main OS), virtual machines (for testing different operating systems), Git and GitHub, REST APIs, and WebSockets (for making a real-time chat app).

ReactNative-Cli (Amateur)

I made an ecommerce app that lets you pay and used the redux toolkit and redux to manage the API and connect it to a real backend.

nodeJs with expressJs and mongodb (Proficient)

I used this as my main backend stack, with MongoDB as the backend.

Languages

Urdu

Fluent

English

Conversational proficiency

RECENT PROJECTS

1- Nexus Store - <https://github.com/codewithahsanzahoor/nexus-store---future> , MERN, TypeScript

- Developed a full-stack **e-commerce application** using **Next.js** for the frontend and **Node.js, Express.js, and MongoDB** for the backend.
- Utilized **TypeScript** to build **type-safe, maintainable, and scalable** application code while improving development reliability.
- Designed and implemented an **Admin Dashboard** for managing **users, products, and e-commerce operations**.
- Implemented centralized **state management** using **Redux Toolkit** for efficient and predictable data handling.
- Developed and integrated **RESTful APIs** to facilitate seamless communication between the **frontend and backend**.

2- Nexus Chat - <https://github.com/codewithahsanzahoor/nexus-chat> , *Real-Time Application - MERN*

- Developed a full-stack **real-time chat application** using **WebSockets** to enable instant, bidirectional communication between users without page refreshes.
- Integrated **Cloudinary** for secure and scalable image storage and media management.
- Implemented **Redux Toolkit** for centralized state management and efficient handling of real-time application data.

3- Book-Store - https://github.com/codewithahsanzahoor/book_store_mern , *MERN Stack*

- Developed a full-stack **E-Book Store** using the **MERN stack** and **TypeScript**, following the **MVC architecture** for scalable and maintainable code.
- Implemented secure **JWT authentication** and integrated **Multer** and **Cloudinary** for image and file upload management.
- Built a responsive frontend using **Tailwind CSS** and managed efficient API communication and server state using **React Query** and **Axios**.

4- Fashion E-commerce App - https://github.com/codewithahsanzahoor/gym_management_system , *React Native CLI & Node.js*

- Built a basic mobile commerce app using **React Native CLI** to learn mobile development fundamentals.
- Implemented product listing and category filtering to understand how to display dynamic data.
- Developed a backend using **Node.js**, **Express**, and **MongoDB** to learn about user registration, login security (JWT), and order management.